

课堂笔记整理 3.31

例 6-1

(a) 由 $Q = CV$, 则

$$Q = 3 \times 10^{-12} \times 20 = 60 \text{ pC}$$

(b) $W = \frac{1}{2} CV^2 = \frac{1}{2} \times 3 \times 10^{-12} \times 400 = 600 \text{ pJ}$

练 6-1

由 $Q = CV$, 则 $V = \frac{Q}{C} = (0.12 \times 10^{-3}) / 4.5 \times 10^{-6} = 26.67 \text{ V}$

则 $W = \frac{Q^2}{2C} = (0.12 \times 10^{-3})^2 / (2 \times 4.5 \times 10^{-6}) = 1.6 \text{ mJ}$

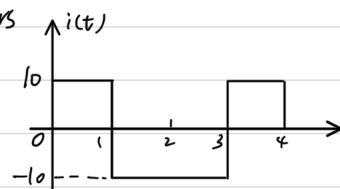
例 6-2

$$i(t) = C \frac{dv}{dt} = 5 \times 10^{-6} \times \frac{d(10 \cos 6000t)}{dt} = -5 \times 10^{-6} \times 10 \times 6000 \times \sin 6000t = -0.3 \sin 6000t \text{ A}$$

例 6-4

$$V(t) = \begin{cases} 50t \text{ V} & 0 < t < 1 \\ (100 - 50t) \text{ V} & 1 < t < 3 \\ (-200 + 50t) \text{ V} & 3 < t < 4 \\ 0 & \text{others} \end{cases}$$

又由 $i = C \frac{dv}{dt}$, 则可得 $i(t) = 200 \times 10^{-6} \times \begin{cases} 50 & 0 < t < 1 \\ -50 & 1 < t < 3 \\ 50 & 3 < t < 4 \\ 0 & \text{others} \end{cases} = \begin{cases} 10 \text{ mA} & 0 < t < 1 \\ -10 \text{ mA} & 1 < t < 3 \\ 10 \text{ mA} & 3 < t < 4 \\ 0 & \text{others} \end{cases}$



练 6-4

$i = \frac{dq}{dt}$, $Q = \int_0^2 i dt = 100 \times 10^{-6} \text{ C}$, $t = 2 \text{ ms}$ 时, $V = \frac{Q}{C} = \frac{100 \times 10^{-6}}{1 \times 10^{-3}} = 100 \text{ mV}$

$t = 2 \text{ ms}$ 至 5 ms 期间, $Q = 100 \times 3 \times 10^{-6} \text{ C}$, $V = \frac{Q}{C} = 400 \text{ mV}$

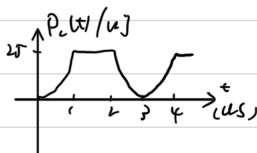
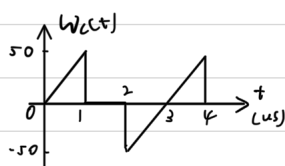
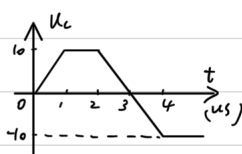
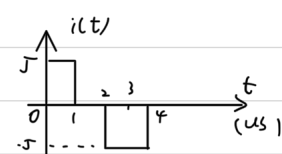
例 6-5

$i = \frac{3}{3+2+4} (6 \text{ mA}) = 2 \text{ mA}$, 则 $V_1 = 2000i = 4 \text{ V}$, $V_2 = 4000i = 8 \text{ V}$

能量为 $W_1 = \frac{1}{2} C_1 V_1^2 = \frac{1}{2} \times (2 \times 10^{-3}) (4)^2 = 16 \text{ mJ}$

$W_2 = \frac{1}{2} C_2 V_2^2 = \frac{1}{2} \times (4 \times 10^{-3}) (8)^2 = 128 \text{ mJ}$

例



例 6-6

$$\frac{20 \times 5}{20+5} = 4 \mu\text{F} \quad 4+6+20=30 \mu\text{F} \quad C_{eq} = \frac{30 \times 60}{30+60} = 20 \mu\text{F}$$

练 6-6

$$\frac{120 \times 60}{120+60} = 40 \mu\text{F} \quad 40+20=60 \mu\text{F} \quad 50+70=120 \mu\text{F} \quad \frac{60 \times 120}{60+120} = 40 \mu\text{F}$$

练 6-7

由 $60 \parallel 30 = \frac{60 \times 30}{60+30} = 20 \mu\text{F}$, 则 $C_{eq} = 40 \parallel 20+20 = 20 \mu\text{F}$

则 $Q = C_{eq} \cdot V = 20 \times 150 = 3000 \mu\text{C}$

$$V_1 = \frac{Q}{C_1} = \frac{3000}{40} = 75 \text{ V}, \quad V_2 = \frac{Q}{C_2} = \frac{3000}{40} = 75 \text{ V}$$

右边并联两个支路, 由于电容相同, 电荷各一半,

$$V_3 = \frac{1500}{60} = 25 \text{ V}, \quad V_4 = V_2 - V_3 = 75 - 25 = 50 \text{ V}$$